UCI Wine Quality Dataset Regression

1. Abstract

Predicting wine (both red wine and white wine) quality from physicochemical properties remains an explorable task due to the local heterogeneity and imbalance of data. This study conducts a systematic comparison of 5 different models, including K-Nearest Neighbors(KNN) Regression, Basis Function/Polynomial Regression(BFR/PR), Multi-Layer Perceptron(MLP), Local Bayesian Regression(LBR) and Symbolic Regression(SR) on UCI Wine Quality Dataset. We incorporate multiple performance metrics, which provides us with a comprehensive understanding of how "fit" the model is(MSE, RMSE, MAE, R^2), and how robust the model is in real-world scenarios(Acc1, ie. accuracy within ±1 quality point). Our findings suggest that wine quality is governed by latent local chemical interactions that global "black-box" models often smooth over due to intense regularization and a focus on population level trend, which positions simpler models, like KNN and LBR, at superior places in this task. While global models, like BFR and MLP, still present decent results, symbolic regression presents limited predictive ability despite interpretability convenience. In conclusion, for small, tabular and locally chemically heterogeneous dataset, simpler models offer a robust balance between complexity and accuracy. Future work should try to address the imbalance of the dataset and further decode the true relationship between physicochemical properties and perceived quality of wines.

2. Introduction

The global wine industry, valued at 330 billion dollars, relies heavily on quality assessment to price products and target consumer preferences. Traditionally, wine quality is evaluated by human experts through sensory analysis, which is subjective, time-consuming and costly. Consequently, inconsistency of quality scores creates a need for objective and data-driven solutions. With the availability of physicochemical data, machine learning(ML) offers a promising avenue to quantify the relationship between chemical ingredients and wine quality.

This project formulates wine quality assessment as a supervised regression problem. The main questions are "How effective physiochemical quality can be used for quality assessment and which model(s) take the lead in this task?" Existing methods normally implement standard models, like simple linear regression and random forest, which may struggle with non-linearity and heterogeneity. Furthermore, many high-performance models operate as "black-box", which offers little insight into which chemical ingredient(s) is the main drive of wine quality.

To address these challenges, we conduct a comprehensive comparative studies of five ML models. Our repository contains classic models covered in the curricula: K-Nearest Neighbors (KNN), Basis Function Regression (BFR), and Multi-Layer Perceptron (MLP), alongside Local Bayesian Regression (LBR) to explore local linearity. Importantly, to boost interpretability and investigate the imbalanced contributions of different properties, we incorporate Symbolic Regression(SR) as our advanced model, which searches for explicit formulas connecting input and output. With the above models selected, we are allowed to test between local predictors and global "black-boxes" and presented with possibility to discover embedded "laws" governing wine quality.

Through this study, we not only explore the predictice performance of different models but also assess better-performing models' robustness across different types of wine(red vs. white). This report details our results in the following order: data pipeline, model setup, main results and future work.

3. Data Pipeline

3.1 Overview and Preprocessing

The UCI Wine Quality dataset contains 6,497 samples(1599 red, 4898 white) with 11 physicochemical features and an integer quality score(0 to 10). We merge the 2 subsets and add a binary "red" indicator as a learnable feature.

Key challenges include:

1. 3:1 white/red ratio, reflecting imbalanced regional and brand production specialization;

2. skewed quality distribution concentrated around [4,7], due to human rating psychology and selection bias .

No missing value is detected.

3.2 Feature Engineering

We retain the original 11 physicochemcal features to ensure a comprehensive analysis of the problem. All continuous features are normalized using "StandardScaler"(zero mean, unit variance). The binary "red" indicator is excluded from normalization.

Although more aggressive feature engineering, such as type-specific normalization, or pairwise interactions, could potentially improve model performance, we end up with employing none of these techniques. Out of deliberation, we choose to prioritize methodological simplicity to focus on our core question: how do different types of models handle locally heterogenous chemical data? We acknowledge that exploring these enhancements may be a valuable direction for future work.

3.3 Handling Imbalance

We conciously retain the original 3:1 white/red ratio for the below reasons:

1. Real-world Scenario: The imbalance reflects production patterns in Vinho Verde region;

2. Model Robustness Test: Evaluating imbalanced data better simulates deployment challenges.

However, we try to mitigate bias by: stratify train/test split; disaggregate evaluation on red/white wines; report ±1 accuracy(ACC1) that is less sensitive compared with MSE.

3.4 Figures and Tables

We present summary table for red, white wines and overall dataset in the appendix, labelled as Table 3, 4, 5, respectively. Additionally, we plot the density for quality score as Figure 1 in appendix.

4. Model Comparison Study

In this section, we compare 5 regression model families on the UCI Wine Quality dataset: K-Nearest Neighbors (KNN) Regression, Local Bayesian Regression (LBR), Basis Function Regression (BFR), Multi-Layer Perceptron (MLP), and Symbolic Regression (SR). These models are selected because they represent different ways to solve the wine quality prediction problem, from simple local methods to more global nonlinear methods. Since the dataset is small, tabular, and contains both red and white wines with different chemical patterns, comparing these different models is meaningful for our task.

4.1 Evaluation Metrics

To evaluate all models fairly, we use the same metrics: mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), coefficient of determination (R^2), and accuracy within ± 1 quality point (ACC1). MSE is used as the main regression metric because it gives larger penalty to larger prediction errors. RMSE is easier to interpret since it is in the same unit as the target variable. MAE shows the average prediction error more directly, while R^2 tells us how much variance in quality is explained by the model. We also report ACC1 because wine quality is an integer score, so predictions within one point of the true quality are still practically acceptable.

4.2 Model Architectures and Hyperparameter Space

All models are implemented in Python(Scikit-learn, PySR), see detail in Table 1.

Table 1: Merged Summary of Model Specifications and Hyperparameter Spaces

Model (Type)	Core Logic & Implementation	Hyperparameter Search Space
KNN	Euclidean distance with inverse weighting to capture local similarity.	$k \in [1,71]$ (odd) & $[72,300]$ (log); *weights*: {uniform, distance}.
LBR	Piecewise linear framework aiming to capture local chemical heterogeneity.	$k \in [25,1038]$; priors $\alpha_1, \lambda_1 \in [10^{-6}, 10^{0}]$; *weights*: {uniform, distance}.
BFR	Polynomial (deg 1–6) and RBF expansion followed by Ridge regression.	$degree \in [1,6]$; $\alpha \in [10^{-4}, 10^{2}]$; RBF centers $\in [\lfloor\sqrt{n}\rfloor, n/5]$.
MLP	Standard MLP with Adam and activation functions ReLU, tanh, and sigmoid. Includes early stopping ($p = 40$) to prevent overfitting.	Layers: [32] to [128, 64, 32]; $\alpha \in [0, 10^{-2}]$; $lr \in [10^{-4}, 10^{-2}]$. Full range of hyperparameters tested in appendix
SR	Evolutionary algorithm (PySR) searching for explicit closed-form expressions.	$max_size \in \{13, 2*13, 3*13\}$; $iterations \in \{20, \lfloor N/40 \rfloor\}$; Population Size $\in$ [20 - 5*13] (3 values evenly spaced) ; basic operators: (+, -, *, /, sqrt, log, exp).

4.3 Tuning Methods and Training Rationale

To endure a robust evaluation of model performance and mitigate the risk of overfitting and underfitting, we first perform a 5-fold cross-validation(CV) on the training dataset(80% of the data). This approach allows each model to be iteratively validated on multiple non-overlapping subsets, ensuring that the selection of hyperparameters present no strong bias and a all-round utilization of the dataset. After choosing the best hyperparameters based on the smallest CV average MSE, we then train our models on the whole training dataset to get the final models. To test for generalizability, we assess the model on the held-out testing dataset(20% of the data, unseen). This tuning and evaluation framework provides us the comparative results that objectively reveals the true predictive ability of our models.

Our tuning process for KNN and LBR reveals a strong presence of local chemical heterogeneity. While both models achieved optimal performancce using inverse-distance weighting, their sensitivity to the neighborhood size(K) differed:

1. **LBR Stability and Rank**: Unlike KNN, the search space for LBR is constrained to ≥ 25 as we try to ensure the local design matrix to be invertible and full rank given our 12 features;

2. **Collapse of LBR**: We spot a collapse of LBR when K is ≤ 35 which often predicts out-of-bounds results. By substituting these with local means, the model is transformed from a linear regressor to a constant predictor. This phenomenon indicates that simpler local averages are more robust in small neighborhoods;

3. **Bayesian Priors**:Tuning α_1 and λ_1(prior shape parameters) enables regularization. However, their impacts are inferior compared to those of K and weights.

Our traing process for BFR aims to balance the sensitivity of polynomials and radial basis functions and potential ill-conditioned problem:

1. **Regularization and Stability**: We utilize a logarithmic range for the regularization intensity to handle unstable cases, particularly for high-degree polynomials;

2. **Polynomial Constraints**: The polynomial degree is contrained to a narrow discrete range(from 1 to 6) to prevent overfitting and high computational cost;

3. **RBF Center Selection**: For RBF training, we view the number of centers as a trainable hyperparameter and select centers from the training dataset directly;

4. **Width Heuristic**: The RBF width is determined by the median pairwise distance heuristic. Distances are estimated from a subset of 500 training data points and spaced logarithmically around the median.

The MLP training explores a moderate variety of network architectures while staying computationally efficient:

1. **Architectural Diversity**: Both single-layer perceptrons(eg. [32]) to multi-layer structures(eg.[128, 64,32]) are trained to assess the effect of different levels of non-linearity;

2. **Activation and Regularization**: ReLU, tanh and sigmoid activations are tested, alongside L2 regularization coefficients from 0 to 10^{-2};

3. **Convergence Guarantee**: Epoch budgets of 2000 and 5000 are tested to ensure convergece while early stop is implemented to reduce overfitting;

4. **Optimization and Efficiency**: Adam optimizer with barch size of 128 is utilized to balance accuracy and speed.

The training for SR is more dynamic as a result of its evolutionary algorithm nature:

1. **Iterations**: The number of iterations for the algorithm is scaled with dataset size since larger datasets may require more iterations to explore the expression space more adequately to capture more complex underlying patterns. ;

2. **Maximum Expression Tree Size**: The maximum sizes is scaled with number of features which is 13. Since complexity of expressions is tied with input dimension (more variables) ;

3. **Population Size**: The population sizes is also scaled with input dimension, simular to maxsize,since more candidate expressions may require larger population size to avoid premature convergence;

4. **Operators**: binary operators and unary operators are explicitly defined, includes basic ones such as addition, multiplication, and sqrt. More complex operators are excluded since they may be too computationally expensive and produce hard to interpret expressions;

5. Main Results

5.1 Final Model Configurations

Table 2: Comprehensive Performance Summary and Optimal Hyperconfigurations

Model Name	Stratified	Optimal Hyperparameters	MSE	R^2	Acc1
KNN	Yes	$k = 23$, weights='distance'	0.368	0.501	96.5%
LBR	Yes	$k = 235$, weights='distancec', α_1 = 1.0 , λ_1 = 10^{-6}	0.351	0.523	97.0%
BFR(Poly)	Yes	degree=2, α=100	0.488	0.337	96.3%
BFR(RBF)	Yes	center=1038, width=0.09 , α=0.215	0.449	0.390	97.1%
MLP	Yes	hidden=[128], act=ReLU, lr=0.001, epoch=2000, α=0	0.452	0.387	96.5%
PySR (SR)	Indifferent	max_size=36, population=40, iterations=20, migration=True	0.612	0.168	93.5%

Note: Acc1 denotes the percentage of predictions within ± 1 of the true quality score. Here, all metrics are evaluated on the testing set.

5.2 Model Selection and Other Findings

For this task, the final model selection is local bayesian regression, which has the leading MSE, R^2 and ACC1. The better performance of local predictor(KNN and LBR) indicates that the physicochemical properties of wine are locally heterogeneous. Additionally, the underperformance of MLP and BFR suggests that nonlinearity is not strongly revealed by the dataset. However, this may be due to the insufficiency of data or the unlearning of dominant drivers of wine quality.

Although MSE varies for different models, we observe that ACC1 is similar for different models, which indicates that all models used are able to predict the wine quality within 1 point range. This finding suggests that all models, based on task-specific requirements, can be potential candidates for real-world evaluation problems as nuanced quality predictions are not always needed.

Additionally, we discover that R^2 scores are relatively low for all models and no model can explain more than 55% of the variation of data. However, as all quality scores provided are integers and we allow prediction results to be unrounded, this may partially explain the low goodness of fit. Furthermore, we believe that the implementation of ACC1 score partly decrease the explainability and importance of R^2.

Moreover, stratification based on red/white wine improves almost all model performances, which reveals that high-quality red wine may be largely different in physicochemical components from white wine. This may also explain that the model trained on the overall training set performs better on red wine quality prediciton than that of white wine. Given the positive effects of stratification, we also predict that stratifying based on quality scores might alleviate the negative impact of the nonuniform quality score distribution of our dataset.

Despite the relatively bad performance of Symnolic Regression, we still gain some insights of this problem from the final expression generated by Symbolic Regression. The expression is 0.39*alcohol+ 5.81(Stratified), which suggests that, just like public opinions, alcohol is the main contributors to a good wine.

In conclusion, the results suggest that moderate-complexity models such as LBR and KNN are the most effective for this dataset, while more flexible nonlinear models such as MLP and SR do not provide a clear advantage. Overall, the best performance comes from models that balance predictive power, stability, and computational efficiency rather than from the most complex model class.

5.3 Supporting Plots

We provide the residual plots for each model in appendix as Figure 2

6. Future Work

While this study demonstrates the viability of physiochemical features for wine quality prediction, several limitations point to the need for future research. The most immediate limitation is the pronounced imbalance between red and white samples. Since the two wine types have distinct physiochemical profiles, the chemical drivers of quality differ between them which cause models trained on the combined dataset to be implicitly biased toward white wine patterns, which contributes to the accuracy differences between the subsets. Future work should explore training type-specific models separately or incorporate wine type as a stronger inductive bias with a more balanced dataset to better capture these chemical patterns.

Naturally, the limitation of extreme quality ratings restricts the models' ability to learn the rare chemical signatures associated with fine or poor wines. A more balanced and diversified dataset, with expert-rated samples across the full quality spectrum, would likely gain meaningful performance gains across all models.

Finally, the prediction framework developed here is not necessarily exclusive to wine. The same physicochemical regression approach could generalise to adjacent domains such as beer or other fermented beverages, where similar measurable properties dictate perceived quality. Extending the scope of this study to a broader dataset spanning multiple beverage types would be a natural and valuable next step.

7. Appendix

For the MLP model, the hyperparameter search tested many hidden-layer architecture, activation function, learning rate, L2 regularization strength, and training budget. The hidden layer architectures were [32], [64], [128], [32,16], [64,32], [128,64], [128,32], [64,16], [128,16], [128,64,32], and [64,32,16]. The activation functions considered were `ReLU`, `tanh`, and `sigmoid`. The learning rates tested were 10^{-4}, 5×10^{-4}, 10^{-3}, 5×10^{-3}, and 10^{-2}; the L2 regularization coefficients were 0, 10^{-5}, 10^{-4}, 10^{-3}, and 10^{-2}; and the maximum iteration budgets were 2000 and 5000. All permutations of such hyperparameters were trained, tested, and compared for a total of 1650 runs.

Table 3: Descriptive Statistics for Red Wine Dataset

	fixed acidity	**volatile acidity**	**citric acid**	**residual sugar**	**chlorides**	**free SO_2**	**total SO_2**	**density**	**pH**	**sulphates**	**alcohol**	**quality**
count	1599.0	1599.0	1599.0	1599.0	1599.0	1599.0	1599.0	1599.0	1599.0	1599.0	1599.0	1599.0
mean	8.32	0.53	0.27	2.54	0.09	15.87	46.47	1.0	3.31	0.66	10.42	5.64
std	1.74	0.18	0.19	1.41	0.05	10.46	32.9	0.0	0.15	0.17	1.07	0.81
min	4.6	0.12	0.0	0.9	0.01	1.0	6.0	0.99	2.74	0.33	8.4	3.0
25%	7.1	0.39	0.09	1.9	0.07	7.0	22.0	1.0	3.21	0.55	9.5	5.0
50%	7.9	0.52	0.26	2.2	0.08	14.0	38.0	1.0	3.31	0.62	10.2	6.0
75%	9.2	0.64	0.42	2.6	0.09	21.0	62.0	1.0	3.4	0.73	11.1	6.0
max	15.9	1.58	1.0	15.5	0.61	72.0	289.0	1.0	4.01	2.0	14.9	8.0

Table 4: Descriptive Statistics for White Wine Dataset

	fixed acidity	**volatile acidity**	**citric acid**	**residual sugar**	**chlorides**	**free SO_2**	**total SO_2**	**density**	**pH**	**sulphates**	**alcohol**	**quality**
count	4898.0	4898.0	4898.0	4898.0	4898.0	4898.0	4898.0	4898.0	4898.0	4898.0	4898.0	4898.0
mean	6.85	0.28	0.33	6.39	0.05	35.31	138.36	0.99	3.19	0.49	10.51	5.88
std	0.84	0.1	0.12	5.07	0.02	17.01	42.5	0.0	0.15	0.11	1.23	0.89
min	3.8	0.08	0.0	0.6	0.01	2.0	9.0	0.99	2.72	0.22	8.0	3.0
25%	6.3	0.21	0.27	1.7	0.04	23.0	108.0	0.99	3.09	0.41	9.5	5.0
50%	6.8	0.26	0.32	5.2	0.04	34.0	134.0	0.99	3.18	0.47	10.4	6.0
75%	7.3	0.32	0.39	9.9	0.05	46.0	167.0	1.0	3.28	0.55	11.4	6.0
max	14.2	1.1	1.66	65.8	0.35	289.0	440.0	1.04	3.82	1.08	14.2	9.0

Table 5: Descriptive Statistics for Overall Dataset

	fixed acidity	**volatile acidity**	**citric acid**	**residual sugar**	**chlorides**	**free SO_2**	**total SO_2**	**density**	**pH**	**sulphates**	**alcohol**	**quality**
count	6497.0	6497.0	6497.0	6497.0	6497.0	6497.0	6497.0	6497.0	6497.0	6497.0	6497.0	6497.0
mean	7.22	0.34	0.32	5.44	0.06	30.53	115.74	0.99	3.22	0.53	10.49	5.82
std	1.3	0.16	0.15	4.76	0.04	17.75	56.52	0.0	0.16	0.15	1.19	0.87
min	3.8	0.08	0.0	0.6	0.01	1.0	6.0	0.99	2.72	0.22	8.0	3.0
25%	6.4	0.23	0.25	1.8	0.04	17.0	77.0	0.99	3.11	0.43	9.5	5.0
50%	7.0	0.29	0.31	3.0	0.05	29.0	118.0	0.99	3.21	0.51	10.3	6.0
75%	7.7	0.4	0.39	8.1	0.06	41.0	156.0	1.0	3.32	0.6	11.3	6.0
max	15.9	1.58	1.66	65.8	0.61	289.0	440.0	1.04	4.01	2.0	14.9	9.0

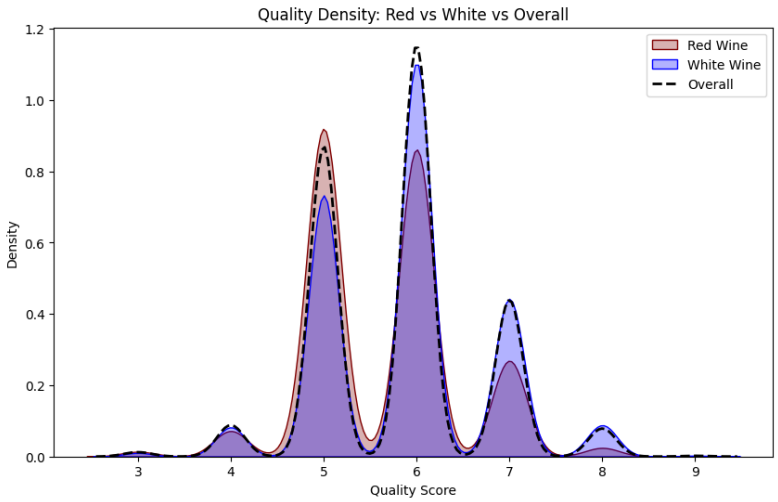
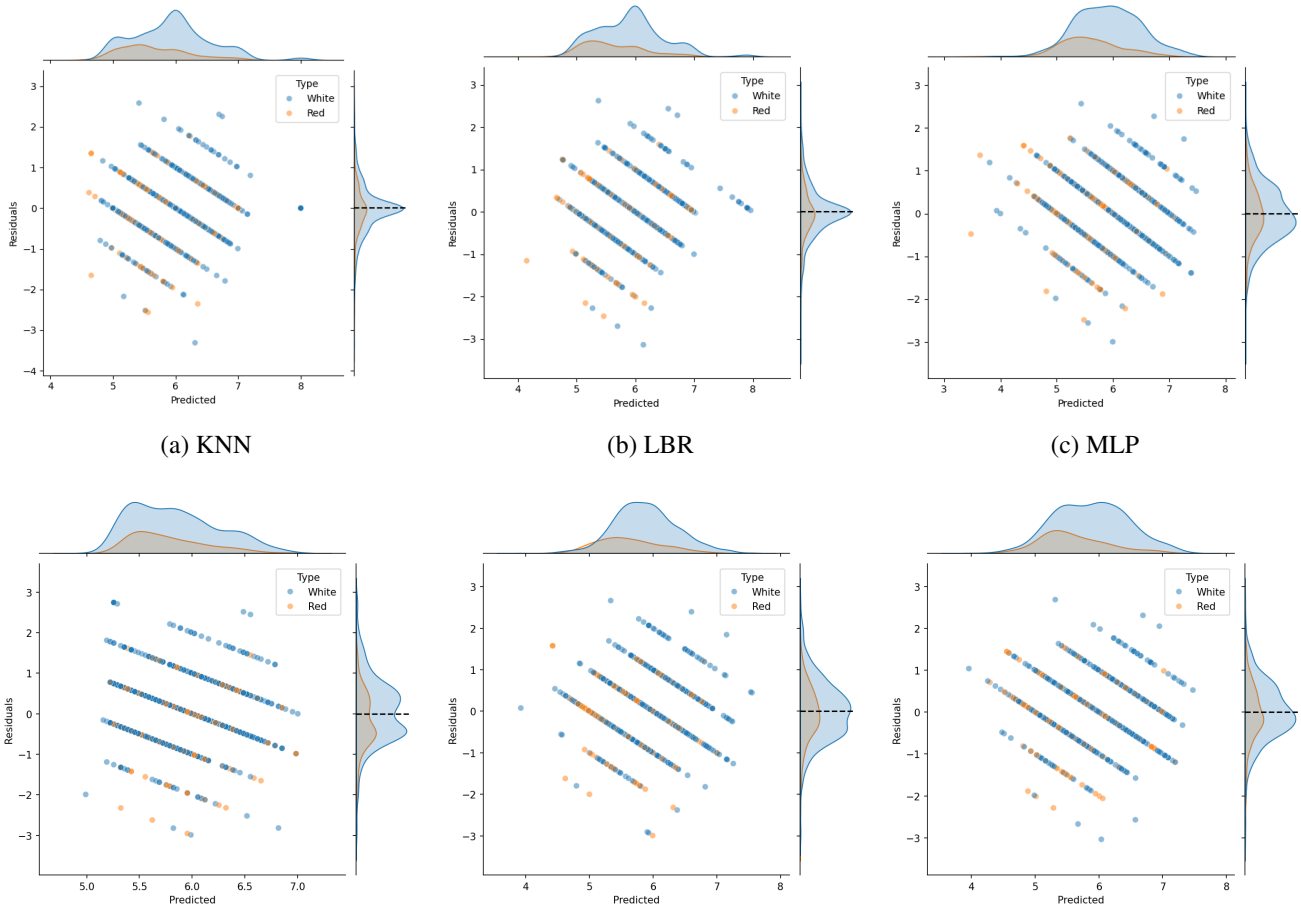


Figure 1: Density plot for quality score, showing imbalanced distribution.



(a) KNN (b) LBR (c) MLP

(d) SR (e) BFR (Poly) (f) BFR (RBF)

Figure 2: Residual plots for all evaluated models, illustrating error distributions across the quality spectrum.